Software Engineering

Lab 3: Component Modelling & Architectural Pattern Selection

56. Peer Skill Exchange & Mentorship Network

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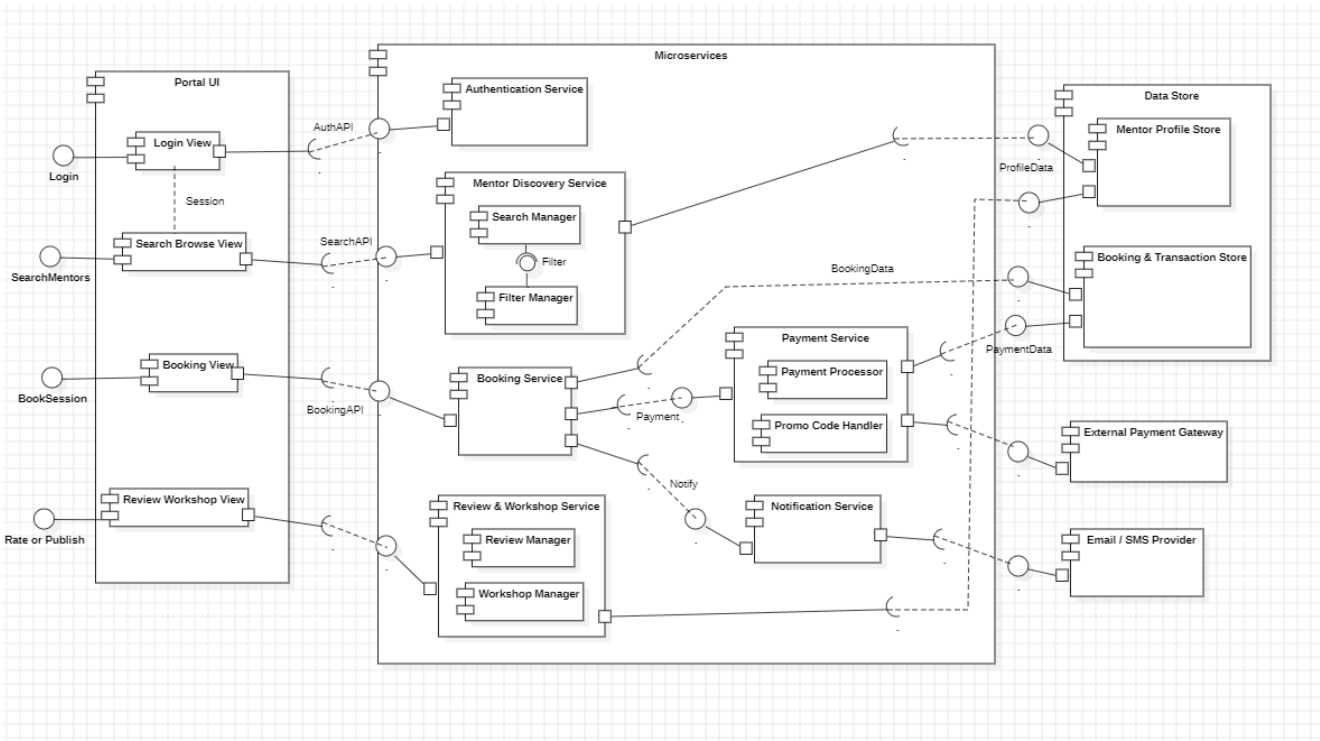
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Deliverables :

1. Component Diagram



2. Written Justification

Architecture Selection
We chose Microservices Architecture for the Peer Skill Exchange & Mentorship Network.

Architectural Choice and Reasons
The system is divided into focused services, including Authentication, Mentor Discovery, Booking, Payment, Review & Workshop, and Notification.

Reasons

1) these services separate the platform's main functions, making each service easier to maintain and update without changing the entire application.

2) services can be scaled independently according to demand—for ex , Mentor Discovery can handle increased search activity without requiring the Booking or Review services to scale at the same time.

Security Advantage
The Authentication Service checks user credentials and tokens, and the API Gateway acts as the main entry point for requests.

Payments are handled through an external payment gateway, reducing the need for the system to handle sensitive card details directly.

Performance Benefit
Mentor search can use a search index or cached profile information to return results faster.

Also, notifications can be sent in the background, so sending an email or SMS does not have to hold up the booking process.